LightGBM for Small-Cap Bankruptcy Prediction in the United States

Word Count: 4353

Abstract

Historically, the stock market has served as the most accessible medium for individuals to invest their capital. To ensure effective investment, individuals should consider a variety of factors to understand the company holistically. These include, but are not limited to, market sentiment, competition analysis, and financial reports. One common investment strategy entails purchasing stock based on market capitalization or company size. Typically, investors gravitate toward large-cap stocks due to their reduced susceptibility to changes in the business cycle, particularly economic downturns (Perez-Quiros & Timmermann, 2000). However, Banz (1981) reveals that small-cap firms produced higher risk-adjusted returns than large-cap firms. These two ideas represent the basis of this study. In the following research paper, a LightGBM model is developed that can help investors effectively harness the returns of small-cap firms while evading the insolvency fears associated with a lower market cap. A Kaggle dataset is leveraged to train the machine learning model to identify patterns associated with bankruptcy. Due to the high-dimensional feature space and quantity of data points, LightGBM is selected as the primary algorithm. The model ultimately predicts with a 95% accuracy score and 60% recall score. Throughout the study, high recall is emphasized while low precision is forgiven: a missed bankruptcy results in 100% capital loss, whereas a false alarm simply results in capital preservation and a missed opportunity.

Keywords: LightGBM, bankruptcy prediction, small-cap, machine learning, gradient boosting, financial distress, SMOTE

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Introduction

Over the past decade, economies globally have endured severe turmoil from the aftermath of the COVID-19 pandemic. Lockdown procedures resulted in heightened unemployment, lower consumer spending, and increased investment fears. These effects led to the collapse of various industries and businesses. Although bankruptcy filings slowed shortly after the immediate dangers of COVID-19 quelled, they have recently begun to re-escalate. According to U.S. Courts (2025), over the trailing-twelve-month period ending March 31, 2025, business bankruptcy filings in the United States rose nearly 14.7%.

Current research points to the idea that these bankruptcies stem from the lingering effects of the 2020 pandemic. Since many businesses stayed afloat by relying on fiscal policy and drawing on credit lines, they were left with insolvencies that contributed to their downfall a few years later. This growth in bankruptcy is likely linked with declining portfolios for investors, too.

Oftentimes, when making investment decisions, the average investor suffers from familiarity bias: the tendency to equate recognizing a brand or company with that investment being "safe." As Huberman (1999) notes, "people invest in the familiar while often ignoring the principles of portfolio theory." Failing to make informed decisions can trigger investment in a company that seems healthy but possesses weak financials underneath. This myopia will result in detrimental investments and capital losses, such as what occurred during the COVID-19 pandemic.

Familiarity bias proves particularly dangerous when applied to small-cap companies—those with a market capitalization between $250M and $2B. These firms operate with less cash and tighter margins, making them highly susceptible to market fluctuations. Consequently, spontaneous economic downturns such as the 2020 pandemic affect these companies the most, fueling bankruptcy. This unpredictability, fused with biased decision-making, can yield significant adverse effects for investment portfolios. Although these firms harbor the most risk, they also possess the potential for the greatest returns. Research suggests that higher returns associated with small-cap stocks are heavily concentrated in firms with a high probability of default (Vassalou & Xing, 2004). In other words, small-cap companies

possess a strong relationship between risk and reward. Therefore, the ability to "remove" risk is highly beneficial to investors seeking the greatest potential reward. By accurately identifying signals of insolvency before they occur, investors can avoid distressed small-caps while investing capital in healthy ones.

Currently, artificial intelligence algorithms represent the primary method for researchers to predict financial distress. Machine learning models first leverage a training set to identify non-linear patterns among financial features that indicate either bankruptcy or financial health. Then, using a testing set, the model applies what it learned to classify unlabeled companies as either failed or healthy. To ensure feasible prediction capabilities, datasets in this realm of financial machine learning contain a mixture of thousands of companies, both healthy and bankrupt. Moreover, due to the natural rarity of bankruptcy, these datasets tend to be highly imbalanced, with significantly more healthy companies compared to bankrupt ones. These high-dimensional feature spaces and massive datasets require efficiency. Consequently, this paper investigates how effectively a LightGBM artificial intelligence model performs in predicting bankruptcy for small-cap companies in the United States.

Literature Review

Researchers have applied machine learning to forecast financial distress for a long time. As a result, this area of artificial intelligence within finance is highly explored, covering early applications of neural networks to support vector machines (SVMs). This review examines the historical shift from single classifiers to ensemble methods, the rise of gradient boosting, and the specific challenges of modeling small and mid-sized enterprises (SMEs) under varying economic conditions.

Foundational research in this space consists primarily of studies pitting individual algorithms against one another to establish baselines. For instance, Min and Lee (2005) show that by mapping input vectors into a high-dimensional feature space, SVMs can solve complex classification problems, outperforming regular logistic regression, multiple discriminant analysis, and backpropagation neural networks. However, as the scope for research widened, the literature shifted more toward ensemble learning—aggregating models to elevate prediction accuracy. Kim

and Kang (2010) highlight that ensemble techniques such as bagging and boosting lower overfitting and improve prediction in volatile financial conditions. This conclusion is confirmed by Lessmann et al. (2015), whose extensive study leveraged international datasets to underscore that ensemble and tree-based models (such as random forests and boosting) routinely outperform more conventional techniques.

Within the ensemble space, gradient boosting has emerged as a strong model for credit risk. Chang et al. (2018) validate that eXtreme Gradient Boosting (XGBoost) yields superior accuracy and Area Under the Curve (AUC) when compared to SVMs in credit risk assessment. Tian et al. (2020) arrive at a similar conclusion. When comparing the effectiveness of common machine learning models—SVMs, decision trees, and multilayer perceptrons—to Gradient Boosting Decision Trees (GBDT) for credit risk assessment, they discover that GBDT demonstrates superior capabilities with the highest accuracy (92.19%), F1-score (91.83%), and AUC (0.97). The statistical effectiveness of gradient boosting in their study posits it as an effective approach to bankruptcy prediction, a field tangential to credit risk.

However, one area of concern is computational efficiency. From a market perspective, small-cap companies compose the majority, making efficiency a critical concern. Models must draw connections between thousands of companies that contain a multitude of features each. Consequently, it is essential to leverage an ML model that can perform well while requiring lower computational intensity. Bentéjac et al. (2020) provide a comparative analysis of modern boosting algorithms, detailing that while XGBoost balances speed and accuracy, LightGBM offers quicker training with comparable predictive power. This advantage becomes especially important when handling high-dimensional data. Moreover, in practice, LightGBM allows researchers and investors to adjust parameters and receive timely outputs.

Although algorithmic architecture is critical, predictive accuracy substantially hinges on firm size and macroeconomic context as well. Malakauskas and Lakštutienė (2021) argued that for SMEs, financial distress “is understood as a significant challenge to a company’s ability to cover liabilities rather than the probability to go bankrupt.” Models must account for specific variables such as age, credit history, and period-at-risk to enhance predictive capabilities. Aker and Karavardar (2023) found that in Turkey’s highly volatile SME economy, decision trees

performed better for short-term prediction, whereas other algorithms such as Naive Bayes excelled over longer horizons.

Even with complex analyses of boosting algorithms and regional studies on SMEs, significant gaps still exist in the literature. Researchers tend to favor XGBoost models and focus mainly on distress prediction in markets outside of the United States (Chang et al., 2018)(Aker & Karavardar, 2023). There is a lack of targeted research evaluating LightGBM's capacity to generalize bankruptcy prediction for U.S. small-cap companies.

Over the lifetime of bankruptcy prediction, researchers have moved scholarly conversation from static statistical models to dynamic, ensemble learning methods. However, the intersection of LightGBM's efficiency and the volatility of U.S. small-cap firms specifically remains underexplored. By testing whether LightGBM can maintain predictive stability for vulnerable U.S. firms over a 19-year time interval, this study addresses the need for a model that is both accurate and robust across economic cycles.

Dataset

Before selecting a dataset, three conditions were outlined for its contents: high dimensionality, an abundant number of small-cap data points, and long-term records. Initially, two approaches were considered: building a custom dataset by compiling company records from SEC filings or manipulating a premade dataset to fit the study's needs. The latter approach was ultimately selected.

The following study leverages Kaggle's *US Company Bankruptcy Prediction Dataset* to train and test the LightGBM model (Singh). This data primarily consists of quantitative information defining each firm's financial position within the broader macroeconomic context. Kaggle is a publicly accessible dataset repository owned by Google. Data scientists, researchers, and organizations contribute their data for others to use. Based on Kaggle's proprietary grading algorithm, the *US Company Bankruptcy Prediction Dataset* received a score of 10.00/10.00, achieving 100% completeness, 100% credibility, and 100% compatibility.

The dataset contains 8,971 unique companies and 78,682 total instances recorded over a 19-year period (1999-2018). Of these 78,682 instances, 93% were marked *alive*, whereas the remaining 7% were marked *failed*. In addition to the company name, status label, and year recorded, each instance contains 18 distinct features detailing the company's financial position, described in Table 1.

Feature	Translation	Description
X1	Current Assets	All the assets of a company that are expected to be sold or used as a result of standard business operations over the next year
X2	Cost of Goods Sold	The total amount a company paid as a cost directly related to the sale of products
X3	Depreciation and Amortization	Depreciation refers to the loss of value of a tangible fixed asset over time (such as property, machinery, buildings, and plant). Amortization refers to the loss of value of intangible assets over time
X4	EBITDA	Earnings before interest, taxes, depreciation, and amortization. A measure of a company's overall financial performance alternative to net income
X5	Inventory	The accounting of items and raw materials that a company either uses in production or sells
X6	Net Income	The overall profitability of a company after all expenses and costs have been deducted from total revenue
X7	Total Receivables	The balance of money due to a firm for goods or services delivered or used but not yet paid for by customers
X8	Market Value	The price of an asset in a marketplace. In our dataset, it refers to the market capitalization since companies are publicly traded in the stock market
X9	Net Sales	The sum of a company's gross sales minus its returns, allowances, and discounts
X10	Total Assets	All the assets, or items of value, a business owns
X11	Total Long-term Debt	A company's loans and other liabilities that will not become due within one year of the balance sheet date
X12	EBIT	Earnings before interest and taxes
X13	Gross Profit	The profit a business makes after subtracting all the costs that are related to manufacturing and selling its products or services
X14	Total Current Liabilities	The sum of accounts payable, accrued liabilities, and taxes such as bonds payable at the end of the year, salaries, and commissions remaining

X15	Retained Earnings	The amount of profit a company has left over after paying all its direct costs, indirect costs, income taxes, and dividends to shareholders
X16	Total Revenue	The amount of income that a business has made from all sales before subtracting expenses. It may include interest and dividends from investments
X17	Total Liabilities	The combined debts and obligations that the company owes to outside parties
X18	Total Operating Expenses	The expense a business incurs through its normal business operations

Table 1: Feature Descriptions (Singh)

Since the dataset was publicly curated, it contained elements that did not fit the parameters of this study. First, the companies recorded in the dataset included all market capitalizations: small-cap, medium-cap, and large-cap. Second, the dataset was highly imbalanced: there were substantially more *alive* states compared to *failed* states (see Figure 1).

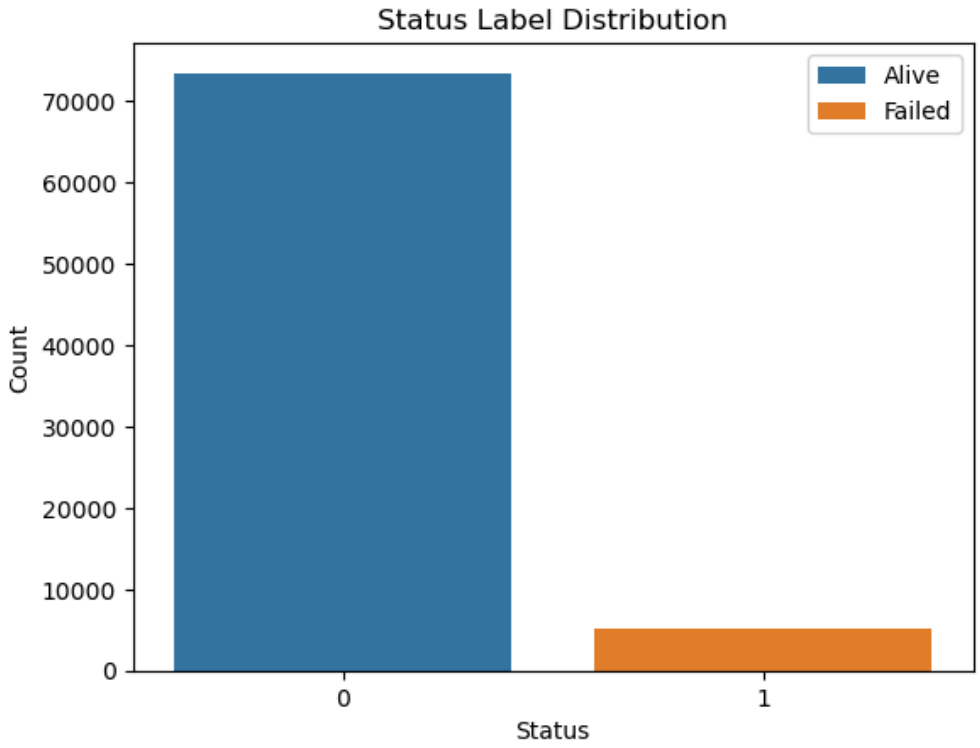


Figure 1: Label Distribution (before filtering)

Although this imbalance reflected the natural rarity of bankruptcy in the real world, it posed a significant risk of overfitting. Due to the high concentration of *alive* instances, the model would become biased toward this majority class, resulting in poor performance in predicting the minority class (*failed*). Third, the curator applied *failed* labels to bankrupt companies at every recorded time point. Earlier records for an eventually bankrupt company were also marked as *failed*, even if the company was financially healthy during that year. These mismatches could confuse the model, leading LightGBM to associate healthy financials with a failed state, thereby diminishing predictive capability. Fourth, the dataset contained high levels of multicollinearity between certain features (see Figure 2). These issues are addressed in the Methodology section.

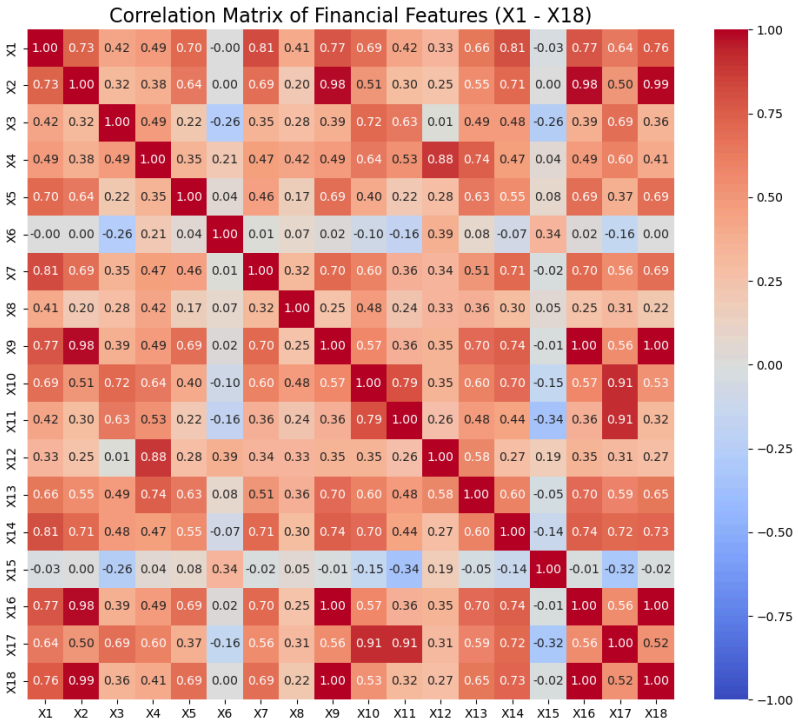


Figure 2: Pearson's Correlation Matrix (before removing multicollinearity)

High multicollinearity would result in certain features dominating the model's predictive process. Since two highly correlated features provide similar information, the model would grant double importance to the underlying quantities attached to those features. The degree of multicollinearity between any two features is quantified by Pearson's correlation coefficient, where a value of 1 represents a direct correlation and a value of -1 represents an inverse correlation. Features such as X2 & X9, X16 & X9, and X16 & X18 possess high

multicollinearity, providing the algorithm with the same information twice. For instance, X2 and X9 likely share high multicollinearity due to their causal relationship: increased net sales suggest a higher amount of goods sold; when more goods are sold, the cost of goods sold increases as well.

Methodology

The research performed in this study follows quantitative experimental methods. The financial indicators applied to train the LightGBM model act as the independent variable, while the predictive outcome of the algorithm represents the dependent variable. Model type and company size serve as the control variables throughout the experiment. The foundation of the experiment relies on a variety of libraries and dependencies described in Table 2.

Library/Dependency	Description
Python	A high-level, multi-purpose programming language used for general-purpose scripting and scientific computing.
Cursor	An AI-native code editor (IDE) based on VS Code, designed for assisted software development and codebase navigation.
Pandas (The pandas development team, 2019)	A library providing high-performance data structures (DataFrames) for tabular data manipulation and analysis.
NumPy (Harris et al., 2020)	A library for numerical computing that supports large, multi-dimensional arrays and high-level mathematical functions.
LightGBM (Shi et al., 2025)	A gradient boosting framework that uses tree-based learning algorithms, optimized for distributed and efficient training.
scikit-learn (Pedregosa et al., 2011)	A machine learning library featuring tools for classification, regression, clustering, and statistical preprocessing.
Optuna (Akiba et al., 2019)	An automatic hyperparameter optimization framework for discovering optimal model parameters through trial-based searching.
seaborn (Waskom, 2021)	A high-level data visualization library based on Matplotlib used for creating informative and complex statistical graphics.
matplotlib (Hunter, 2007)	A low-level library for creating static, animated, and interactive visualizations and plots.

Table 2: Library and Dependency Descriptions

Preprocessing

After downloading and importing the *US Company Bankruptcy Prediction Dataset*, several preprocessing steps were carried out to address the issues described in the Dataset section.

Market Capitalization Filtering

The dataset incorporated companies of all market capitalizations, not exclusively small-caps, as required by this study. Since feature X8 denoted the market value of each instance, a function was executed to isolate instances falling within the small-cap range of $250M–$2B and remove all others. After this preprocessing step, 2,474 unique small-cap companies and just over 20,000 instances remained (see Figure 3).

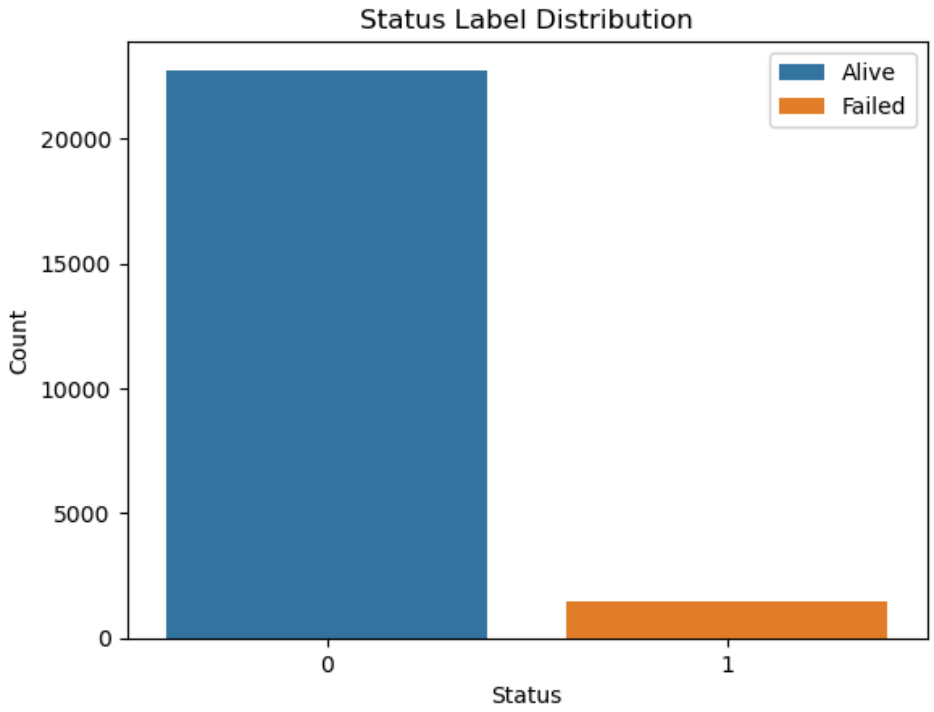


Figure 3: Label Distribution (after filtering)

Removing Multicollinearity

To prevent LightGBM from granting double importance to underlying financial quantities, highly correlated features were identified using Pearson's correlation matrix. Features with a multicollinearity score exceeding a threshold of 0.85 were flagged. To determine which feature from each correlated pair to remove, a baseline LightGBM model was created to measure each feature's information gain—its contribution to the model's predictions (see Figure 4).

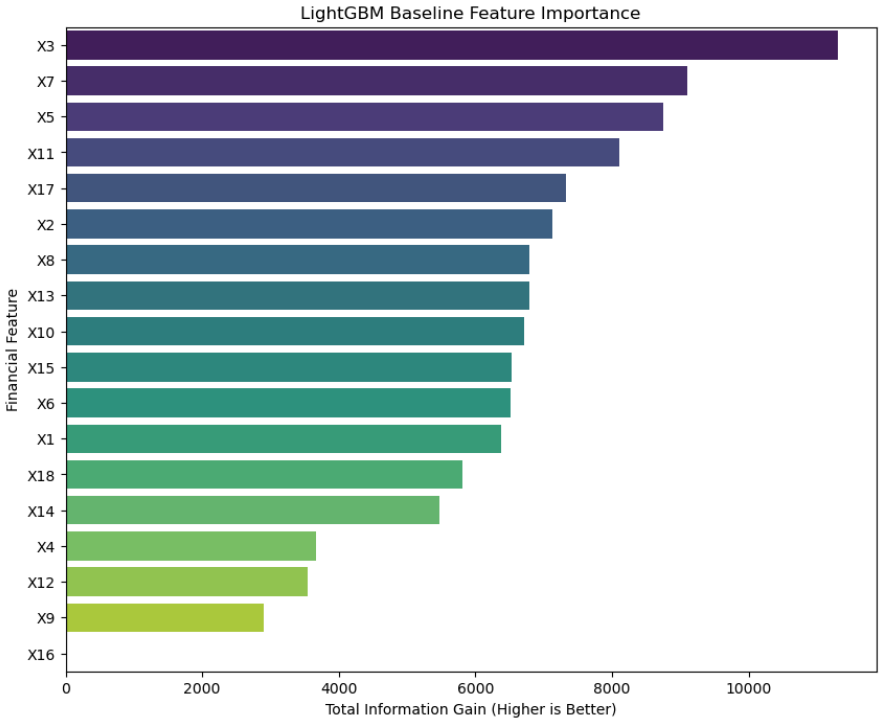


Figure 4: Feature Importance Chart (before removing multicollinearity)

The lower-importance feature from each correlated pair was removed. For instance, between X2 and X9 (multicollinearity = 0.98), X9 was removed because it contributed lower information gain (≈3,000) compared to X2 (≈7,500). Similarly, between X10 and X17 (multicollinearity = 0.91), X10 was removed because X17 provided greater information gain (≈7,750 vs. ≈7,000). This process was repeated for all correlated feature pairs, resulting in a dataset where no two features shared a multicollinearity score above 0.85 (see Figure 5).

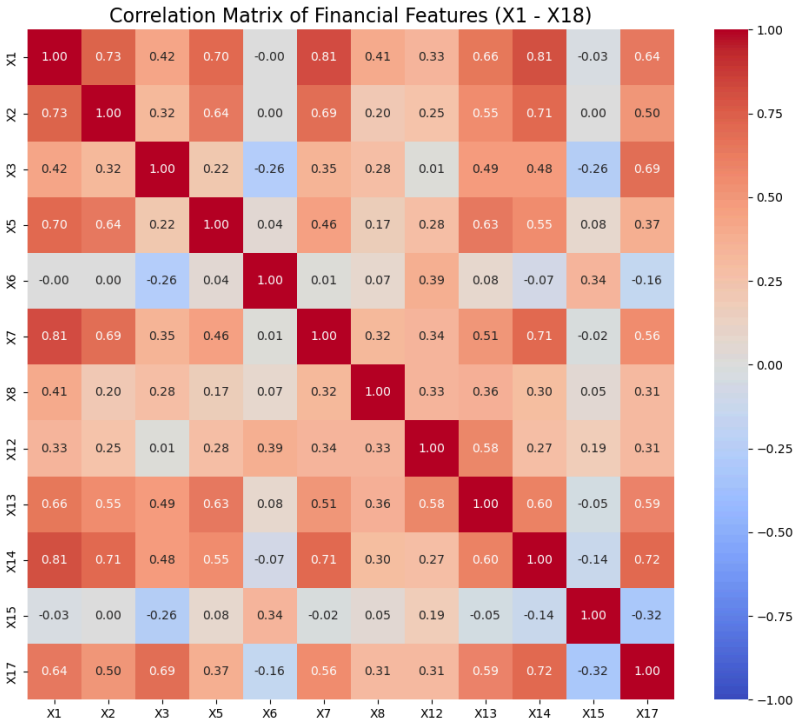


Figure 5: Pearson's Correlation Matrix (after removing multicollinearity)

Label Correction

To address the curator's oversight of applying *failed* labels to every recorded instance of an eventually bankrupt company, the dataset was parsed to identify all *failed* instances. These were organized by company, and each instance prior to the final year of bankruptcy was relabeled as *alive*. Only the instance corresponding to the year the company actually filed for bankruptcy retained the *failed* label. This ensures LightGBM does not associate financially healthy data with a failed state.

Class Imbalance: SMOTE

Severe class imbalance, as present in the *US Company Bankruptcy Prediction Dataset*, can result in predictions biased toward the majority class. As Guo et al. (2008) note, "standard machine learning algorithms tend to be overwhelmed by the majority class and ignore the minority class since traditional classifiers seek accurate performance over a full range of instances." In response, the Synthetic Minority Over-sampling Technique (SMOTE) was applied, increasing the number of minority-class instances to match the majority class. Unlike standard over-sampling techniques that simply duplicate minority instances, SMOTE operates within the

feature-space. Specifically, SMOTE "selects points along the line segments joining k minority class nearest neighbors to create synthetic data" (Chawla et al., 2002). This process is depicted in Figure 6.

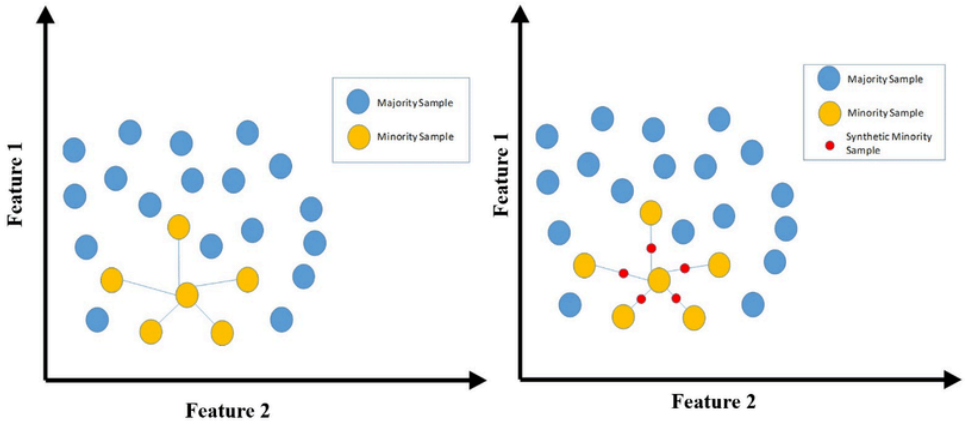


Figure 6: SMOTE Visual Representation (Vijayvargiya et al., 2021)

For each new synthetic instance, SMOTE chose a point on the direct link between two minority features, thereby creating data that reflects the financials of a bankrupt company without repeating existing observations. This approach minimizes overfitting and increases generalizability on the testing set. After applying SMOTE, the dataset consisted of equally balanced classes: 20,941 *alive* data points and 20,941 *failed* data points, displayed in Figure 7.

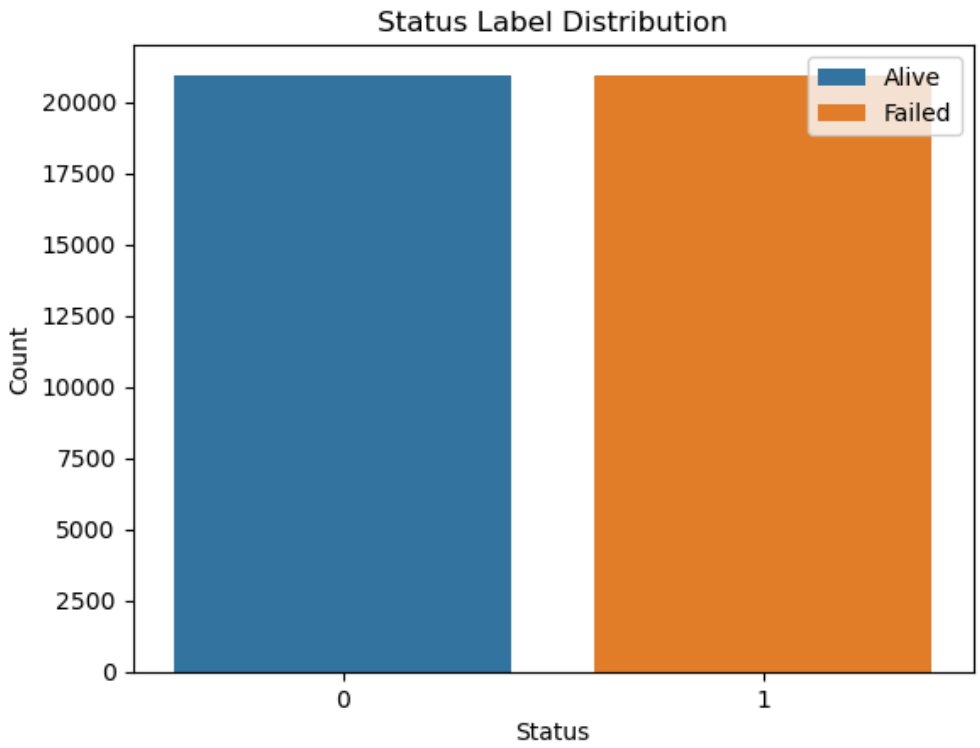


Figure 7: Label Distribution (after filtering and SMOTE)

Model Training and Evaluation

Following preprocessing, the LightGBM model was trained using a temporal 80%/20% train-test split: data from 1999 to 2015 for training and from 2015 to 2018 for testing. This temporal split prevents data leakage, where the model could use future information to predict past outcomes. Time series cross-validation (TSCV) was subsequently executed on the training set rather than traditional *k*-fold cross-validation, which shuffles data before creating folds and risks exposing the model to future data. TSCV folds data chronologically within the training set, ensuring temporal order is preserved.

A baseline LightGBM model was then trained to reveal shortcomings in the prediction process and to serve as the control group for subsequent hyperparameter tuning with Optuna (Akiba et al., 2019). Hyperparameter tuning selects optimal configuration settings—such as num_leaves (maximum number of nodes per tree) and max_depth (maximum distance from root to leaf node)—to augment predictive capabilities. Optuna automates this search through Bayesian optimization (see Figure 8).

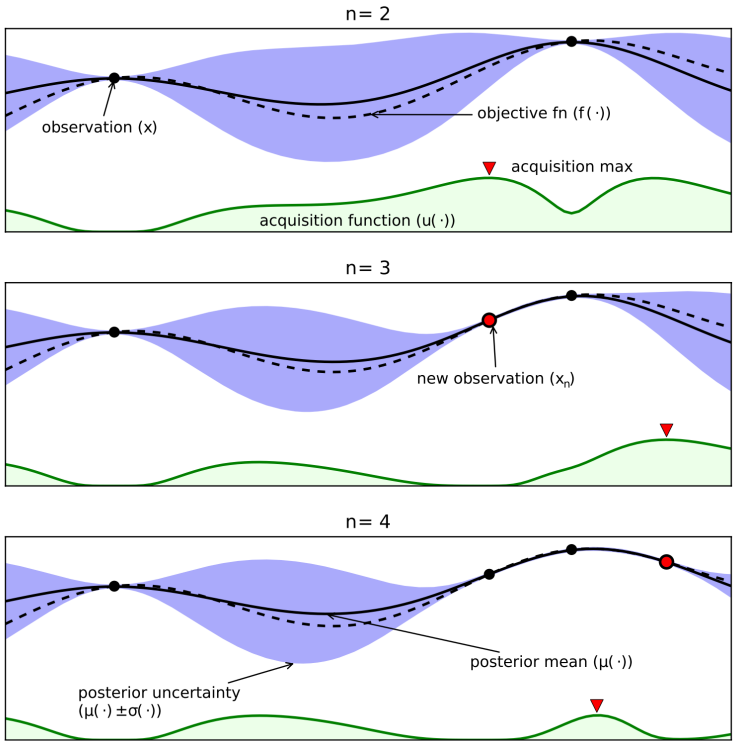


Figure 8: Illustration of Bayesian Optimization (Shahriari et al., 2016)

It begins with a random set of parameters, analyzes performance, constructs an acquisition function to identify promising regions, and iteratively selects parameters that maximize the acquisition function until the optimal hyperparameter set is discovered.

Evaluation Metrics

Both untuned (baseline) and tuned LightGBM models were evaluated using confusion matrices (see Figure 9 for example), accuracy, precision, F1-score, and recall. Within this study, *failed* states represent the positive class (1) and *alive* states represent the negative class (0). The relevant prediction classification terms are defined as follows:

- **TP:** LightGBM correctly predicted a *failed* company as *failed.*
- **TN:** LightGBM correctly predicted an *alive* company as *alive.*
- **FP:** LightGBM incorrectly predicted an *alive* company as *failed.*

- **FN:** LightGBM incorrectly predicted a *failed* company as *alive.*

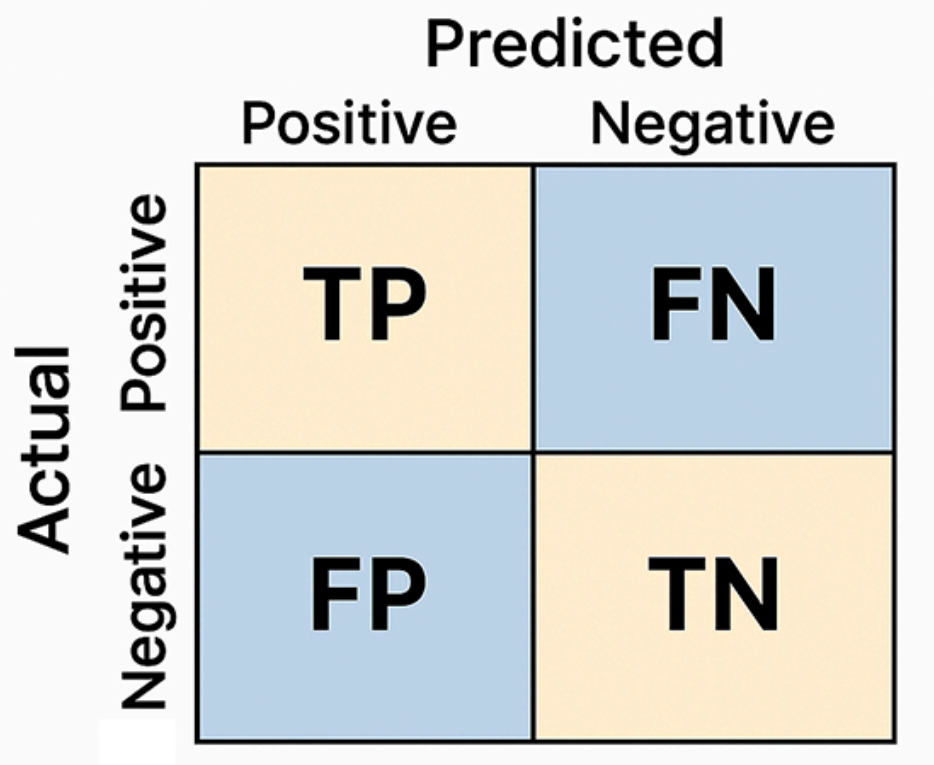


Figure 9: Annotated Confusion Matrix

The four evaluation metrics are mathematically defined as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

Throughout the evaluation, additional emphasis is placed on high recall, while low precision is forgiven. A missed bankruptcy (false negative) can result in 100% capital loss,

whereas a false alarm (false positive) results only in capital preservation and a missed opportunity.

Results and Analysis

Feature Selection Observations

One notable observation during the methodology was that X16 (Total Revenue) provided zero information gain to the original LightGBM model. Although this does not directly answer the research question, it reveals an interesting property of LightGBM and tree-based models in general. LightGBM robustly handles multicollinearity within its decision-making process because it follows leaf-wise tree growth (see Figure 10): at every split, LightGBM selects the leaf node whose split reduces global loss (or maximizes information gain) the most, until the desired max_depth or num_leaves is reached.

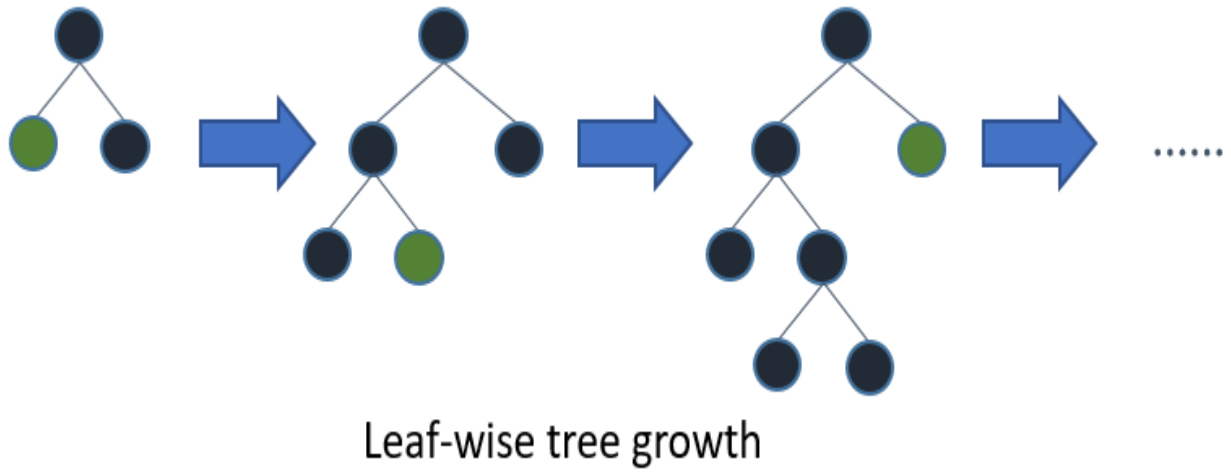


Figure 10: Illustration of Leaf-wise Tree Growth (Liu et al., 2017)

When two features share high multicollinearity, LightGBM inherently selects the one with greater information gain, effectively handling duplicate information within its growth process. Consequently, at every split, X16 provided lower information gain than its competitor feature, resulting in a cumulative information gain of zero. Contrary to expectations, total revenue provided no value to predicting bankruptcy.

Although LightGBM can handle multicollinearity internally, highly correlated features were still removed to increase interpretability and reduce computing requirements.

Model Performance

The results for both the baseline and tuned LightGBM models are displayed in Table 3. Confusion matrix details are as follows (see Figures 11 & 12): the baseline model produced TP = 15, TN = 3,046, FP = 89, FN = 15; the tuned model produced TP = 18, TN = 2,998, FP = 137, FN = 12.

LightGBM Results

Model	Accuracy	Precision	F1-Score	Recall
Baseline LightGBM	0.96714	0.14423	0.22388	0.50000
Tuned LightGBM	0.95292	0.11612	0.19459	0.60000

Table 3: LightGBM Results (Baseline & Tuned)

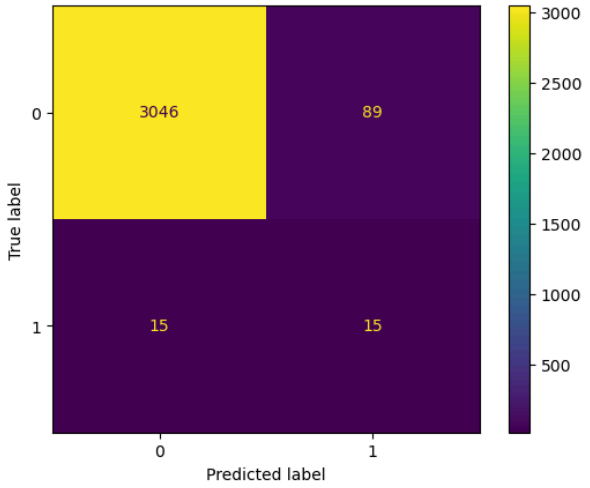


Figure 11: Baseline LightGBM Confusion Matrix

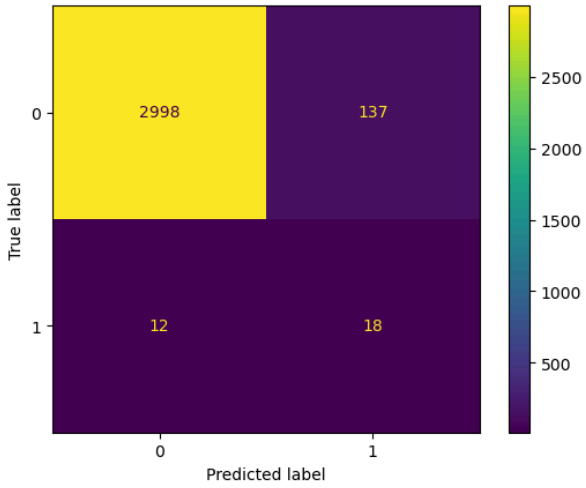


Figure 12: Tuned LightGBM Confusion Matrix

Accuracy score plays a limited role in this context. Due to the class imbalance in the testing set and in real-world data, a model that predicted every instance as *alive* would achieve approximately 99% accuracy, since 99% of instances are *alive*. Greater emphasis is therefore placed on the remaining metrics.

As shown in Table 3, the baseline LightGBM outperformed the tuned model across all metrics except recall. The tuned model correctly identified three additional bankruptcies (a 10% improvement in recall) at the expense of 48 additional false positives compared to the baseline model.

Despite the decline in overall precision and F1-score, the tuned LightGBM is considered superior for small-cap bankruptcy prediction when accounting for the asymmetric risks in financial contexts. Low precision is forgiven because false positives carry relatively limited consequences for investors: even if a company does not ultimately fail, financials that mirror those of a failing company provide a genuine warning signal. The investor's only loss is the opportunity cost of not investing. High recall is prioritized because it maximizes the number of correctly identified bankrupt companies, thereby decreasing the probability that investors place capital into critically distressed firms. Although increased recall comes at the cost of more false positives, this tradeoff is justified: higher recall reduces the chance of complete capital loss, whereas additional false positives merely reduce the chance for gains on investment.

Feature Importance

The principal features contributing to LightGBM's predictions were X17 (Total Liabilities), X8 (Market Value), X5 (Inventory), and X7 (Total Receivables). Notably, the tuned model extracted significantly greater information gain from these features compared to the baseline model, as illustrated in Figures 13 & 14 below.

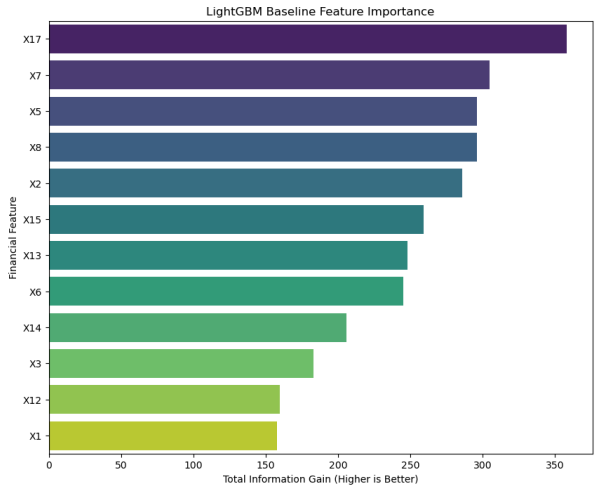


Figure 13: Baseline LightGBM Feature Importance

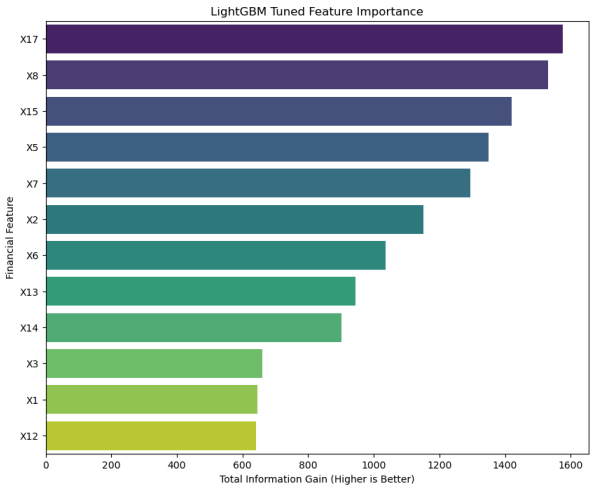


Figure 14: Tuned LightGBM Feature Importance

One intriguing observation was the prominence of market value (X8) in the prediction process. Given that market capitalization was isolated to the small-cap range ($250M–$2B), a low information gain from this feature was expected. However, the wide dollar range encompassed by small-cap companies means that variance in firm size was sufficient for the model to discern a meaningful relationship between market value and bankruptcy likelihood.

Conclusion

With the current state of these models, LightGBM should be used cautiously to forecast small-cap bankruptcy in the United States. It performs slightly above average, successfully predicting the majority of firms' status labels. Despite growth in recall score, LightGBM still fails to detect bankruptcy in 40% of instances, which can have serious effects on investment portfolios. If the LightGBM classifications are supplemented with additional data sources and features, predictive capabilities could be substantially improved.

The limited classification performance can be attributed to several methodological constraints. Most notably, the dataset was not curated specifically for small-cap companies; filtering by market capitalization reduced the number of *failed* instances available for training. Additionally, applying SMOTE to create a large number of synthetic instances from such a small positive class likely introduced some degree of overfitting that diminished overall predictive capability.

Future research could benefit from introducing new features that provide LightGBM with richer information. For example, a feature capturing contemporary macroeconomic health would help the model differentiate the financial profile of a company failing during a recession from one failing during an expansionary period. Researchers could also incorporate non-quantitative features, such as public sentiment, to reveal additional dimensions of each company's condition. Together, these additions could significantly boost LightGBM's performance in financial distress contexts.

Ultimately, this research establishes LightGBM as an above-average model for predicting bankruptcy among small-cap companies in the United States. By addressing the shortcomings identified herein, LightGBM has the potential to become an effective and robust tool for bankruptcy prediction within this context.

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